LOGIC BUILDING WITH LOOPS

(BEFORE STARTING DSA)

🎯 **Goal: Master loops, iteration, and dry-run thinking.**
Topics covered: while, do-while, for, break, continue, mathematical series, pattern printing.
Target Questions: 125+

Phase 1 : While Loop

1. Print all numbers from 1 to 10 using a loop.
2. Print numbers from 10 down to 1 in reverse order.
3. Print all even numbers between 1 and 100.
4. Print all odd numbers between 1 and 100.
5. Print the multiplication table of a given number from n × 1 to n × 10.
6. Calculate and print the sum of the first n natural numbers.
7. Calculate the sum of all even numbers from 1 up to n.
8. Calculate the sum of all odd numbers from 1 up to n.
9. Calculate and print the factorial of a given number.
10. Find and print the product of all digits of a given number.
11. Count and print the total number of digits in a given number.
12. Reverse the given number and print the reversed value.
13. Check whether the given number is a palindrome.
14. Find and print the sum of digits of the given number.
15. Check whether the given number is an Armstrong number.
16. Check whether the given number is a Perfect number.
17. Print all prime numbers between 1 and 100.
18. Check whether the given number is a prime number.
19. Print the Fibonacci series up to n terms.
20. Find and print the sum of the Fibonacci series up to n terms.
21. Print the square of each number from 1 to n.
22. Print the cube of each number from 1 to n.
23. Print all numbers between a and b that are divisible by 7.
24. Print all factors of the given number.
25. Find and print the sum of all factors of the given number.
26. Find the HCF (Highest Common Factor) of two given numbers.
27. Find the LCM (Least Common Multiple) of two given numbers.
28. Find the smallest digit in the given number.
29. Find the largest digit in the given number.

Phase 2 : do - While Loop

1. Print all numbers from 1 to 10.
2. Print the multiplication table of a given number.
3. Keep taking numbers from the user until 0 is entered, then print the sum of all entered numbers.
4. Keep taking numbers from the user until 0 is entered, then print the largest number among all inputs.
5. Count and print the number of digits in the given number.
6. Reverse the given number and print the reversed value.
7. Check whether the given number is a palindrome.
8. Check whether the given number is an Armstrong number.
9. Calculate and print the factorial of the given number.
10. Print the Fibonacci series up to the required number of terms.
11. Find the HCF (Highest Common Factor) of the given numbers.
12. Create a menu-driven program that allows the user to choose and perform different operations.
13. Keep taking numbers from the user until a negative number is entered, then count how many positive numbers were entered.
14. Find and print the sum of digits of the given number.
15. Calculate and print the sum of even digits and the sum of odd digits of the given number separately.

Phase 3 : For Loop

1. Print all numbers from 1 to 10.
2. Print numbers from 10 down to 1 in reverse order.
3. Print all even numbers between 1 and 100.
4. Print all odd numbers between 1 and 100.
5. Print the multiplication table of a given number.
6. Calculate and print the factorial of a given number.
7. Calculate and print the factorial of every number from 1 to n.
8. Print all prime numbers between 1 and 100.
9. Check whether the given number is a prime number.
10. Print the Fibonacci series up to the required number of terms.
11. Find and print the sum of the Fibonacci series.
12. Print all factors of the given number.
13. Find and print the sum of all factors of the given number.
14. Find the HCF (Highest Common Factor) of the given numbers.
15. Find the LCM (Least Common Multiple) of the given numbers.
16. Print the square of each number from 1 to n.
17. Print the cube of each number from 1 to n.
18. Print all numbers between a and b that are divisible by 7.

19. Find and print the sum of the first n natural numbers.
20. Find and print the sum of all even numbers from 1 up to n.
21. Find and print the sum of all odd numbers from 1 up to n.

Phase 4 : Nested Loop Logic

1. Print the multiplication tables for all numbers from 1 to 10.
2. Print all possible pairs (i, j) where both i and j range from 1 to n.
3. For every number from 1 to n, count and print the total number of its factors.
4. Print all prime numbers up to n using nested loop checking.
5. Print the Fibonacci pattern row by row, where each row prints the next Fibonacci numbers.
6. Generate and print a number triangle pattern using nested loops.
7. Print a matrix, then calculate and display the sum of each row and the sum of each column.
8. Print all Pythagorean triplets whose values are less than or equal to n.

Phase 5 : Break / Continue Logic

1. Print numbers from 1 to 100, and stop the loop as soon as a number divisible by 17 is encountered.
2. Print numbers from 1 to 100, but skip all numbers that are divisible by 5 and continue printing the rest.
3. Take 5 numbers as input, skip any number that is 0 using continue, and calculate the sum of the remaining numbers.
4. Search for a specific number in a list of inputs, and terminate the loop immediately when the number is found.
5. Keep taking numbers from the user and print them until a negative number appears, then stop the loop.
6. Skip all odd numbers and print only the even numbers.
7. Continuously add numbers in a loop and stop the loop when the sum becomes greater than 100.

Phase 6 : Mathematical Series

1. Find and print the sum of the first n natural numbers.
2. Find and print the sum of the first n even numbers.
3. Find and print the sum of the first n odd numbers.
4. Print the first n terms of an arithmetic progression for the given first term and common difference.
5. Print the first n terms of a geometric progression for the given first term and common ratio.
6. Print the Fibonacci series up to the required number of terms.
7. Find and print the sum of the Fibonacci series up to the required number of terms.
8. Calculate and print the value of the series: $1^2 + 2^2 + 3^2 + \dots + n^2$.
9. Calculate and print the value of the series: $1^3 + 2^3 + 3^3 + \dots + n^3$.
10. Calculate and print the value of the series: $1 + 1/2 + 1/3 + \dots + 1/n$.
11. Print the series of powers of two: $1 + 2 + 4 + 8 + \dots + 2^n$.
12. Calculate and print the value of the series: $1! + 2! + 3! + \dots + n!$.

13. Calculate and print the value of the series: $1 + x + x^2 + x^3 + \dots + x^n$.

14. Calculate and print the value of the series: $x - x^2/2! + x^3/3! - x^4/4! + \dots$.

15. Check whether the given number is a Strong number, where the number equals the sum of factorials of its digits.

Phase 7 : Mixed Logical Loop Problems

1. Print all numbers between 1 and 100 whose sum of digits is even.
2. Count Total numbers between 1 & 500 are divisible by 7 but not divisible by 5.
3. Print all palindrome numbers between 1 and 500.
4. Print all numbers from 1 to 100 whose sum of digits is a multiple of 3.
5. Print all numbers from 1 to n whose binary representation contains an even number of 1s.
6. Print a pattern where the i-th row prints the value $i \times i$.
7. Find & print the sum of odd digits & the sum of even digits of the given number.
8. Print all Armstrong numbers between 1 and 1000.
9. Print all Perfect numbers between 1 and 1000.
10. Find the number between 1 and n that has the maximum digit sum, and print that number along with its digit sum.

Phase 8 : Star & Number Pattern Printing

1. Print a Single Star (*)
2. Print Four Stars (****)
3. Print n Stars on Same Line
4. Print Square of Stars (n x n Stars)
5. Print an Increasing Triangle of Stars
6. Print a Right-Aligned Triangle of Stars
7. Print Stars in Even Numbers (2, 4, 6, 8, 10)
8. Print Stars in Odd Numbers (1, 3, 5, 7, 9)

```
*
***
*****
*******
*********
```

9. Print a Centered Pyramid of Stars
10. Print Stars and Spaces Alternating (where b = blank space):

```
bbbb*
bbb*b*
bb*b*b*
b*b*b*b*
*b*b*b*b*
```

11. Print Numbers in an Increasing Sequence:

    1
    12
    123
    1234
    12345

12. Print Repeated Numbers per Row:

    1
    22
    333
    4444
    55555

13. Print Continuous Number Triangle (Floyd's Triangle):

    1
    2 3
    4 5 6
    7 8 9 10
    11 12 13 14 15

14. Print Continuous Single-Digit Loop Pattern:

    1
    2 3
    4 5 6
    7 8 9 0
    1 2 3 4 5
    6 7 8 9 0 1
    2 3 4 5 6 7 8

15. Print Binary Alternate Pattern:

    1
    0 1
    0 1 0
    1 0 1 0
    1 0 1 0 1

16. Print Sequential Alphabet Triangle:

    A
    B C
    D E F
    G H I J
    K L M N O

17. Print Repeated Alphabet Triangle:

    A
    B B
    C C C
    D D D D
    E E E E E

18. Print Column Alphabet Triangle:

A
A B
A B C
A B C D
A B C D E

19. Print Continuous Alphabet Pyramid:

A
BCD
EFGHI
JKLMNOP
QRSTUVWXY

20. Print Number Half Pyramid:

1
12
123
1234
12345

21. Print Palindromic Number Pyramid:

1
121
12321
1234321
123454321

22. Print Diamond / Hourglass Star Pattern

23. Print Half Diamond Star Pattern

24. Print Centered Diamond Star Pattern

25. Print Decreasing-Increasing Number Pyramid Pattern:

5
545
54345
5432345
543212345